

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a PySpark job to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Databricks Cluster for DLT
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, pyspark

Source Data Details:

- Dataset Name: customerdata
- Location: `/Volumes/gen\_ai\_poc\_databrickscOE/sdlc\_wizard/customerdata`
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: orderdata
- Location: `/Volumes/gen\_ai\_poc\_databrickscOE/sdlc\_wizard/orderdata`
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: customer
- Location: Delta table in catalog `gen\_ai\_poc\_databrickscOE` and schema `sdlc\_wizard`
- Format: Delta
- Description: Customer data in Delta format
- Output Name: order

- Location: Delta table in catalog `gen\_ai\_poc\_databrickscoe` and schema `sdlc\_wizard`
- Format: Delta
- Description: Order data in Delta format

Job Flow / Pipeline Stages:

- Read customer CSV data into a Delta table named "customer"
- Read order CSV data into a Delta table named "order"
- Add a new column "TotalAmount" to the "order" table

Data Transformations / Business Logic:

- Step: Read customer CSV data
- Description: Read customer CSV data into a Delta table
- Transformation Logic: Use PySpark to read CSV data and write to Delta table
- Step: Read order CSV data
- Description: Read order CSV data into a Delta table
- Transformation Logic: Use PySpark to read CSV data and write to Delta table
- Step: Add TotalAmount column
- Description: Add a new column "TotalAmount" to the "order" table
- Transformation Logic: Use PySpark to calculate TotalAmount as the product of PricePerUnit and Qty

Error Handling and Logging:

- Log errors to a log file
- Handle null and duplicate records

TR-DLT-002

Related Functional Requirement(s): FR-DLT-002

Objective: Implement data quality checks to remove null and duplicate records from customer and order Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: customer and order Delta tables

Target Data Details: cleaned customer and order Delta tables

Job Flow / Pipeline Stages:

- Remove null records from customer and order Delta tables
- Remove duplicate records from customer and order Delta tables

Data Transformations / Business Logic:

- Step: Remove null records

- Description: Remove null records from customer and order Delta tables
- Transformation Logic: Use PySpark to filter out null records
- Step: Remove duplicate records
- Description: Remove duplicate records from customer and order Delta tables
- Transformation Logic: Use PySpark to drop duplicates

Error Handling and Logging:

- Log errors to a log file
- Handle any exceptions during data quality checks

TR-DLT-003

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a PySpark job to create and load the ordersummary table with joined customer and order data using SCD Type 2.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: customer and order Delta tables

Target Data Details: ordersummary Delta table

Job Flow / Pipeline Stages:

- Create the ordersummary table if it does not exist
- Join customer and order Delta tables on CustId
- Load the joined data into the ordersummary table using SCD Type 2

Data Transformations / Business Logic:

- Step: Join customer and order data
- Description: Join customer and order Delta tables on CustId
- Transformation Logic: Use PySpark to join the two tables
- Step: Load data into ordersummary table
- Description: Load the joined data into the ordersummary table using SCD Type 2
- Transformation Logic: Use PySpark to implement SCD Type 2 logic

Error Handling and Logging:

- Log errors to a log file
- Handle any exceptions during data loading

TR-DLT-004

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a PySpark job to update the ordersummary table when there are changes to the customer data.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: customer Delta table

Target Data Details: ordersummary Delta table

Job Flow / Pipeline Stages:

- Identify changes to the customer data
- Update the ordersummary table by making old records inactive and new records active

Data Transformations / Business Logic:

- Step: Identify changes to customer data
- Description: Identify changes to the customer data
- Transformation Logic: Use PySpark to compare old and new customer data
- Step: Update ordersummary table
- Description: Update the ordersummary table by making old records inactive and new records active
- Transformation Logic: Use PySpark to implement SCD Type 2 logic

Error Handling and Logging:

- Log errors to a log file
- Handle any exceptions during data updates

TR-DLT-005

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a PySpark job to create and load the customeraggregatespend table with aggregated data from the ordersummary table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: ordersummary Delta table

Target Data Details: customeraggregatespend Delta table

Job Flow / Pipeline Stages:

- Create the customeraggregatespend table if it does not exist
- Aggregate the TotalAmount column from the ordersummary table
- Load the aggregated data into the customeraggregatespend table

Data Transformations / Business Logic:

- Step: Aggregate TotalAmount
- Description: Aggregate the TotalAmount column from the ordersummary table

- Transformation Logic: Use PySpark to group by Name and Date and sum TotalAmount
- Step: Load data into customeraggregatespend table
- Description: Load the aggregated data into the customeraggregatespend table
- Transformation Logic: Use PySpark to write to Delta table

Error Handling and Logging:

- Log errors to a log file
- Handle any exceptions during data loading

TR-DLT-006

Related Functional Requirement(s): FR-DLT-006

Objective: Implement Delta Live Tables (DLT) to execute the above steps.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: N/A

Target Data Details: N/A

Job Flow / Pipeline Stages:

- Implement DLT to execute steps 1 to 5 above

Data Transformations / Business Logic:

- Step: Implement DLT
- Description: Implement Delta Live Tables to execute the above steps
- Transformation Logic: Use DLT to create a pipeline that executes the above steps

Error Handling and Logging:

- Log errors to a log file
- Handle any exceptions during DLT execution